Dr. Deva Nanda Panda

bm25mtech14001@iith.ac.in | devajonny2001@gmail.com | +91 9573442066 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.64
Kendriya Vidyalaya, Picket	Class XII (CBSE)	2018	87.00%
Kendriya Vidyalaya, Picket	Class X (CBSE)	2016	10.00%

PROJECTS

HIFU Robotic Positioning & Thermal Mapping System

- Built a MATLAB-controlled robotic positioning system using an igus 4-axis arm for automated HIFU transducer targeting
- Developed an Arduino-based thermocouple acquisition system for real-time sonication temperature logging
- Validated in agarose phantoms and ex vivo goat tissue; ongoing ex vivo sonication of resected human meningioma and schwannoma specimens at NIMHANS

HIFU Acoustic & Thermal Simulation

- Developed 2D/3D k-Wave simulations across multiple HIFU transducers, modeling pressure, intensity, and heat deposition
- Evaluated thermal dose using CEM43 criteria and quantified focal spot dimensions
- Extended pipeline to CT-derived skull geometry for post-craniectomy focused ultrasound targeting

In-vitro Evaluation of Dental Matrix Systems

- Compared 4 dental matrix systems on anatomical contact tightness and restoration contour accuracy across 200 typodont teeth
- First author; published in J Conserv Dent Endod 2025

SCHOLASTIC ACHIEVEMENTS

Academic Excellence Award - IIT Hyderabad (2026)

Awarded for securing the **highest academic performance in M.Tech. Biomedical Engineering (First Year)**

ICMR Short Term Studentship (STS) 2022 - Indian Council of Medical Research

- Completed a 2-month ICMR research project on dental matrix systems (Ref. 2022-11504) under Dr. Mamta Kaushik; report certified satisfactory.

3rd Place - Poster Presentation, UDAAN 2023

Secured 3rd place for presenting and defending a scientific research poster at UDAAN 2023, organized by the Indian Dental Association.

PUBLICATION

Assessment of anatomical contacts and contours of class II restorations using four different dental matrix systems: An in vitro comparative analysis. J Conserv Dent Endod 2025;28:449-55

SKILLS

Ultrasound & HIFU: HIFU, Ultrasound, k-Wave, Acoustic & Thermal Modeling, CEM43 analysis

Biomedical: Medical Imaging, MRI, Biomedical Sensors, Health Informatics

Robotics: Robotics Fundamentals, Robotic Systems

Programming: Python, MATLAB, PyTorch

AI/ML: Machine Learning, Deep Learning

POSITIONS OF RESPONSIBILITY

Placement Coordinator | OCS, IIT Hyderabad | Apr 2026 - Present

Managed placement-related activities, facilitated communication between students and recruiters, and supported the coordination and execution of recruitment drives and campus placement events.

Technical Secretary | Army College of Dental Sciences | Apr 2022 - Feb 2024

Integrated the hospital management system with paper-based workflows across 8 Departments. Moved student communication, grievance, and complaint handling onto a digital system serving 400+ Students. Managed the auditorium's technical operations.

CERTIFICATIONS

Python for Everybody - University of Michigan - Issued November 2025 (Credential ID: UO81USR5BRD7)

Machine Learning Specialization - Stanford University - Issued December 2025 (Credential ID: LT9GQ2MUUHIH)

Generative AI for Healthcare - Google - Issued February 2026 (Credential ID: FUUZ42IHUXUJ)

Health Informatics for Healthcare Professionals - Northeastern University - Issued March 2026 (Credential ID: P41W3BIXL00K)

Basics of Robotics - Siemens - Issued September 2026 (Credential ID: 4A2C4GAY0Y4R)

EXTRACURRICULAR

Lead a 100-member international alliance in an online strategy game across 11+ countries to set strategy, resolve conflicts, and coordinate activity across time zones

Participation Award in University Basketball competition